KAIZEN MATHS

Lesson starter

Poisson Distribution

Formula and Basic Probabilities

Name _____

S1

Use the Poisson formula to calculate each exact probability.

If $X \sim Po(5)$, find $P(X = 1)$.

S2

Write the cumulative probability as a sum of exact Poisson probabilities.

If $X \sim Po(5)$, find $P(X \leq 2)$.

S3

Use a complement where it makes the probability shorter.

If $X \sim Po(3)$, find $P(X \geq 1)$.

S4

Use the key facts for a Poisson distribution.

If $X \sim Po(9)$, state $E(X)$, $Var(X)$, and the standard deviation.

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Show your working

KAIZEN MATHS

Exit ticket

Poisson Distribution

Formula and Basic Probabilities

Name

E1

Use the Poisson formula to calculate each exact probability.

If X ~ Po(2), find P(X = 3).

E2

Write the cumulative probability as a sum of exact Poisson probabilities.

If X ~ Po(6), find P(X ≤ 4).

E3

Use a complement where it makes the probability shorter.

If X ~ Po(5), find P(X ≥ 3).

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Exit ticket

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Use a complement where it makes the probability shorter.

If X ~ Po(5), find P(X ≥ 3).

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Use a complement where it makes the probability shorter.

If X ~ Po(5), find P(X ≥ 3).

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If X ~ Po(2), find P(X = 3).

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Write the cumulative probability as a sum of exact Poisson probabilities.

If X ~ Po(6), find P(X ≤ 4).

E3

Use a complement where it makes the probability shorter.

If X ~ Po(5), find P(X ≥ 3).

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Teacher answers & working

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S1

Use the Poisson formula to calculate each exact probability.

If X ~ Po(5), find P(X = 1).

Answer:

$\mathrm{P}(\mathrm{X} = 1) = 0.0337.$

Working:

For X ~ Po(lambda), P(X = r) = $\frac{e^{-lambda} lambda^{r}}{r!}$.

Here lambda = 5 and r = 1.

$\mathrm{P}(\mathrm{X} = 1) = \frac{e^{-5} 5^{1}}{1!}$.

$\mathrm{P}(\mathrm{X} = 1) = 0.0337.$

S2

Write the cumulative probability as a sum of exact Poisson probabilities.

If X ~ Po(5), find P(X ≤ 2).

Answer:

$\mathrm{P}(\mathrm{X} \leq 2) = 0.1247.$

Working:

A cumulative probability is a sum of exact probabilities.

$\mathrm{P}(\mathrm{X} \leq 2) = \mathrm{P}(\mathrm{X} = 0) + \mathrm{P}(\mathrm{X} = 1) + \mathrm{P}(\mathrm{X} = 2).$

$\mathrm{P}(\mathrm{X} \leq 2) = 0.0067 + 0.0337 + 0.0842.$

$\mathrm{P}(\mathrm{X} \leq 2) = 0.1247.$

S3

Use a complement where it makes the probability shorter.

If X ~ Po(3), find P(X ≥ 1).

Answer:

$\mathrm{P}(\mathrm{X} \geq 1) = 0.9502.$

Working:

Use the complement because X ≥ 1 has many possible values.

$\mathrm{P}(\mathrm{X} \geq 1) = 1 - \mathrm{P}(\mathrm{X} \leq 0).$

$\mathrm{P}(\mathrm{X} \leq 0) = \mathrm{P}(\mathrm{X} = 0).$

$\mathrm{P}(\mathrm{X} \leq 0) = 0.0498.$

$\mathrm{P}(\mathrm{X} \geq 1) = 1 - 0.0498 = 0.9502.$

Teacher answers & working

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S4

Use the key facts for a Poisson distribution.

If X ~ Po(9), state E(X), Var(X), and the standard deviation.

Answer:

$\mathrm{E}(\mathrm{X}) = 9$, $\mathrm{Var}(\mathrm{X}) = 9$, $\mathrm{SD} = 3.000$.

Working:

For a Poisson distribution, the mean is lambda and the variance is also lambda.

E(X) = lambda = 9.

Var(X) = lambda = 9.

Standard deviation = sqrt(lambda) = sqrt(9) = 3.000.

E1

Use the Poisson formula to calculate each exact probability.

If X ~ Po(2), find P(X = 3).

Answer:

$\mathrm{P}(\mathrm{X} = 3) = 0.1804$.

Working:

For X ~ Po(lambda), P(X = r) = $\frac{e^{-lambda}lambda^{r}}{r!}$.

Here lambda = 2 and r = 3.

$\mathrm{P}(\mathrm{X} = 3) = \frac{e^{-2}2^{3}}{3!}$.

$\mathrm{P}(\mathrm{X} = 3) = 0.1804$.

E2

Write the cumulative probability as a sum of exact Poisson probabilities.

If X ~ Po(6), find P(X ≤ 4).

Answer:

$\mathrm{P}(\mathrm{X} \leq 4) = 0.2851$.

Working:

A cumulative probability is a sum of exact probabilities.

$\mathrm{P}(\mathrm{X} \leq 4) = \mathrm{P}(\mathrm{X} = 0) + \mathrm{P}(\mathrm{X} = 1) + \mathrm{P}(\mathrm{X} = 2) + \mathrm{P}(\mathrm{X} = 3) + \mathrm{P}(\mathrm{X} = 4)$.

$\mathrm{P}(\mathrm{X} \leq 4) = 0.0025 + 0.0149 + 0.0446 + 0.0892 + 0.1339$.

$\mathrm{P}(\mathrm{X} \leq 4) = 0.2851$.

Teacher answers & working

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E3

Use a complement where it makes the probability shorter.

If X ~ Po(5), find P(X ≥ 3).

Answer:

$P(X \geq 3) = 0.8753$.

Working:

Use the complement because X ≥ 3 has many possible values.

$P(X \geq 3) = 1 - P(X \leq 2)$.

$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$.

$P(X \leq 2) = 0.1247$.

$P(X \geq 3) = 1 - 0.1247 = 0.8753$.